METRIC

DOD-B-24531(SH)
23 February 1976

MILITARY SPECIFICATION

BATTERY, STORAGE, SILVER-ZINC ALKALINE (EMERGENCY);

GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the rechargeable silver-zinc alkaline storage batteries used for emergency onboard deep submergence vehicles.

1.2 Classification. Silver-zinc alkaline storage batteries shall be of the types specified (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date on invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-576 - Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges (Plate, Bar, Sheet, and Strip).
QQ-S-365 - Silver Plating (Electrodeposited); General Requirements for.

MILITARY

DOD-B-24531/1 - Battery, Storage, Silver-Zinc Alkaline (Emergency), For DSV: Type 25 V - 12 Ah (METRIC).
DOD-B-24531/2 - Battery, Storage, Silver-Zinc Alkaline (Emergency), For DSRV: Type 25 V - 70 Ah (METRIC).
DOD-B-24531/3 - Battery, Storage, Silver-Zinc Alkaline (Emergency), For Trieste: Type 12 V - 100 Ah (METRIC).
DOD-B-24531/4 - Battery, Storage, Silver-Zinc Alkaline (Emergency), For Trieste: Type 24 V - 200 Ah (METRIC).

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INCORPORATED, AGENT

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington DC 20036.)

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.1.1).

3.2 Electrical requirements.

3.2.1 Electrical rating. The electrical rating shall be as specified in the applicable specification sheet.

3.2.2 Capacity retention on stand. The battery shall be capable of retaining 90 percent of its rated capacity after a charged stand of 30 days in an ambient 25°C ± 3°C.

3.2.3 Charge acceptance. The battery shall be capable of accepting a full capacity charge in 16 hours or less.

3.2.4 Life. The battery shall be capable of meeting the life requirement specified in the applicable specification sheet before its capacity drops to 70 percent of the rated capacity over a one year period after its activation date.

3.3 Physical requirements.

3.3.1 Dimensions. The dimensions shall be as specified in the applicable specification sheet.

3.3.2 Mass. The mass shall be as specified in the applicable specification sheet.

3.3.3 Cell case and cover. The cell case and cover shall be molded out of a thermoplastic material resistant to potassium hydroxide (KOH). This material shall withstand temperatures ranging from 0°C to 70°C.

3.3.4 Cell vent valve. Each cell shall be provided with a removable vent valve.

3.3.5 Terminal hardware. Terminals and associated hardware shall be silver-plated in accordance with QQ-S-365, type III, grade B. The dimensions shall be as specified in the applicable specification sheet.

3.3.6 Intercell connectors. The intercell connectors shall be made of copper in accordance with QQ-C-576 and silver-plated in accordance with QQ-S-365, type III, grade B.

3.3.7 Electrolyte. The electrolyte shall be aqueous KOH of concentration not less than 42 percent KOH by mass, with no additives, with a potassium carbonate content not to exceed 2 percent by mass. The electrolyte shall be a part of a filling kit to be supplied with each battery delivered. As a minimum, the kit shall contain the following:

(a) Sufficient vials of premeasured electrolyte for filling each cell.
(b) Pair of tweezers.
(c) Absorbent cotton.

3.3.8 Cell design characteristics.

3.3.8.1 Electrode construction. The positive electrodes shall have smooth edges and corners rounded to a radius of at least 3 millimeters (mm).

3.3.8.2 Electrode grids. Silver shall be used for the positive and negative grids supporting the active material. The negative grid shall have smooth edges and corners rounded to a radius of at least 3 mm.

3.3.8.3 Gas evolution. The gassing rate on charged open circuit stand at 25°C ± 3°C shall be not greater than an average of 4 mm^3/hour/gram of negative active material converted into zinc metal over a 6-hour period, when measured at hourly intervals.

3.4 Environmental requirements.

3.4.1 Temperature. The battery shall be capable of meeting the electrical requirements as specified in the applicable specification sheet at an ambient temperature from 10°C to 50°C.

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3.4.2 Angle of operation. The battery shall not spill any electrolyte when it is tilted 45 degrees from the vertical.

3.5 Hardware requirements. The total number of batteries and cells to be delivered shall be as specified (see 6.1.1). Each battery specified in the contract or order shall be delivered with a filling kit and electrolyte (see 3.3.7).

3.6 Mercury. The service manual and shipping instructions shall refer to the content (amount and form) of the mercury compound in each cell, if used. The amount of mercury shall not exceed 2 percent by weight of the active material of the zinc electrode. If mercury is used, each battery shall be provided with a label as follows:

CAUTION - MERCURY

CONTAINS MERCURY COMPOUND
IN NEGATIVE ELECTRODES

NAME: _____ AMOUNT_____ grams

USE APPROVED BY NAVSEA

IN THE EVENT OF A SHORT CIRCUIT RESULTING IN SMOKE OR FIRE, ISOLATE THE AREA AND NOTIFY THE COMMANDING OFFICER OR OFFICER-IN-CHARGE, MEDICAL OFFICER, AND QUALITY ASSURANCE DEPARTMENT.

3.7 Identification.

3.7.1 Battery. In addition to the label specified in 3.6, each battery shall be provided with a permanently mounted label identifying the following, as a minimum:

(a) Caution: Contains potassium hydroxide when filled.
(b) Battery type designation and serial number.
(c) Contract or order number.
(d) Date of delivery.
(e) Manufacturer's name.

3.7.2 Terminals. The battery terminals shall be legibly marked "POS", "P", or "+" for positive and "NEG", "N", or "-" for the negative.

3.8 Technical data. The supplier shall prepare technical data in accordance with the data ordering documents included in the contract or order and as specified in 3.8.1 and 3.8.2. The supplier shall use the International System of Units (SI) in all technical data. For threaded hardware designation, the Unified Thread Standard shall be used.

3.8.1 Drawings. The supplier shall prepare engineering drawings (includes documents referenced therein) for review by the Naval Ship Engineering Center (NAVSEC) (see 3.9) in accordance with the data ordering document included in the contract or order (see 6.1.2).

3.8.2 Service manual. The supplier shall prepare a service manual to cover each battery described in the individual specification sheets. The manual shall be in accordance with the data ordering document included in the contract or order (see 6.1.2), and in accordance with the appendix to this specification.

3.9 Design review. Design review meetings with Government representatives shall be held prior to the release of any drawings to vendors, or production, or finalization of the service manual.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program plan. The supplier shall provide and maintain a quality program plan in accordance with the data ordering document included in the contract or order (see 6.1.2).

4.2 Sampling for quality conformance inspection. Five cells from the entire production lot of cells of the same type shall be randomly selected for quality conformance inspection specified in 4.3.

4.3 Quality conformance inspection.

4.3.1 Inspection in dry condition.

4.3.1.1 Examination. The entire production lot of cells of the same type shall be examined for workmanship and for conformance to the physical requirements of this specification (see 3.3) and the applicable specification sheet.

4.3.1.2 Air-pressure test. All cells of the entire production lot shall be submitted to a differential air-pressure of 14 kilopascals (kPa) for 5 minutes with no more than 0.7 kPa drop to determine the adequacy of case-to-cover seal and terminal seals.

4.3.2 Inspection in wet conditions. The sample cells only shall be filled with electrolyte, formed, and tested as specified in 4.3.2.1 through 4.3.2.3.

4.3.2.1 Capacity test. Each sample cell shall be discharged at 25°C ± 3°C at the lowest rate specified in the applicable specification sheet to the specified cut-off voltage. Each cell individually shall meet the capacity and voltage requirements specified in the applicable specification sheet.

4.3.2.2 High rate test. Each sample cell shall be discharged at the highest rate and pulse rate specified in the applicable specification sheet. Each cell shall meet the voltage requirements.

4.3.2.3 Gassing rate. The gassing rate of each sample cell shall be measured as specified in 3.3.8.3 and shall meet the requirements of 3.3.8.3.

4.3.2.4 Failure procedure. In case of failure, the tests specified in 4.3.2.1 and 4.3.2.2 may be repeated once.

4.3.2.5 Disposition of sample cells. After completion of tests, all sample cells shall be discharged and shipped to a Government laboratory for further testing as specified in 4.4.

4.4 Government laboratory tests. The sample cells of the same type shall be series-connected to form a battery and shall be tested as specified in 4.3.2.1 through 4.3.2.3, 4.4.1 and 4.4.2.

4.4.1 Capacity retention. After recharge, the 5-cell battery shall be placed on charged stand in an ambient temperature of 25°C ± 3°C for 30 days, then discharged at the lowest rate to the specified cut-off voltage specified in the applicable specification sheet. The 5-cell battery shall be recharged and given one more identical cycle. Both cycles shall meet the requirements specified in 3.2.2.

4.4.2 Life test. The 5-cell battery shall be placed on the following regime to establish conformance with 3.2.4:

(a) Three cycles per week: discharge at the 4-hour rate for 2 hours, recharge as recommended in the service manual.
(b) Charged stand over the week-end: (record open circuit voltage after end of last charge and before start of next discharge).
(c) One capacity discharge after every 3 months at the lowest rate to the specified cut-off voltage in accordance with the applicable specification sheet. Record average voltage and capacity.
(d) Continue the sequence until the 5-cell battery fails to deliver 2 hours at the 4-hour rate.

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4.5 Inspection of preparation for delivery. The preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be as specified in 5.1.1 through 5.1.3.

5.1.1 Level C preservation-packaging. Preservation-packaging shall afford protection against damage, leakage, and deterioration during shipment from the supply source to the receiving activity for early installation. The manufacturer's normal packaging procedure may be used when such meets the requirements of this level.

5.1.2 Packing. Batteries shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest rate. Containers, packing, and method of shipment shall comply with Uniform Freight or National Motor Freight Classification rules and regulations. Shipping containers shall contain batteries of identical size, and shall be vented.

5.1.3 Marking. In addition to any special markings required by the contract order, interior packages (unit and intermediate) and exterior shipping containers shall be marked and labeled in accordance with MIL-STD-129 as applicable to the levels of packaging and packing specified. In addition, each exterior shipping container shall be labeled as specified in 3.6.

6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification and applicable specification sheet.
(b) Type required (see 3.1).
(c) Address for shipment of batteries.
(d) Number of batteries and cells to be delivered (see 3.5).
(e) Special marking required (see 3.7).
(f) Address for shipment of sample cells to a Government laboratory (see 4.3.2.5).

6.1.2 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited data item description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

	Specification paragraph	Data requirement	Service	Applicable DID	Option
(a)	3.8.1	Drawings	SH	UDI-E-23174	Categories B, G, and H, form 2, type II
(b)	3.8.2	Service manual	SH	DI-M-6157	------
(c)	4.1.1	Quality program plan	SH	UDI-R-23743	------

(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

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6.2 Guarantee. Procurement documents should contain the following life guarantee to be applied by the supplier to the batteries under this specification:

Guarantee: "Notwithstanding inspection and acceptance by the Government of supplies furnished under the contract or any provisions of this contract concerning the conclusiveness thereof, the supplier guarantees the performance of the battery in terms of service operating conditions, months of service or total ampere-hour output, as specified in the applicable specification sheet, before the capacity drops to 70 percent of the rated capacity. The period of the guarantee shall begin on the date the battery is filled with electrolyte but not later than 24 months after delivery. If the battery fails to meet these requirements, the supplier will replace the battery at no cost to the Government."

Preparing activity:
Navy - SH
(Project 6140-N422)

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APPENDIX

SERVICE MANUAL FORMAT FOR EMERGENCY SILVER-ZINC BATTERIES

10. SCOPE

10.1 This appendix covers the format for the preparation of a service manual for the emergency silver-zinc batteries. The format shall be as follows:

Chapter (Cover pages)		Page
	TITLE PAGE	1
	APPROVAL AND PROCUREMENT RECORD PAGE	2
	VALIDATION OF PERFORMANCE	3
	TABLE OF CONTENTS	4

1. GENERAL INFORMATION

103 Description of cells and battery plus following statement of format: The format of this manual is condensed from that of the military specification for procuring the main batteries for this deep submergence vehicle. Thus non-applicable paragraphs and chapters have been deleted but the paragraph and chapter numbers for any given topic are consistent from one manual to the other.

2. FUNCTIONAL DESCRIPTION

A. ELECTROCHEMISTRY

201 General
202 Reactions

3. PREPARATION

A. SHIPMENT

301 Dry shipment
302 Wet shipment

B. STORAGE

311 Dry
312 Wet charged
313 Wet discharged

C. PREPARATION FOR INSTALLATION

321 Safety precautions
322 Filling cells
323 Formation (if necessary)

4. OPERATION

A. GENERAL

401 Safety precautions

B. CHARGING

411 General characteristics
412 State of charge
413 Normal charge
414 Overcharge
420 Voltage limits
421 When to charge
422 Ventilation
423 Temperature limits

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C. DISCHARGING

431 General characteristics
432 Discharge rate
433 Voltage limits
434 Ventilation
435 Temperature limits
436 Capacity test discharge

D. BATTERY RECORDS

442 Battery monthly report

5. MAINTENANCE

A. REQUIRED ACTION

501 General
502 Capacity maintenance
503 Preventive maintenance
504 Maintenance schedule

B. STANDARDS

511 Voltage
512 Capacity
515 Meters
517 Cell seals
519 General cleanliness

D. SUPPORT SHIP AND SHORE BASED ACTION

531 Voltage
535 Meters
537 Cell seals
538 Battery seals
539 General cleanliness

E. RECOMMENDATIONS

541 General
543 Booster charge
544 Temporary storage
545 Low temperature storage

6. TROUBLE SHOOTING

A. FAILURE MODES

601 Capacity loss
602 Internal shorting

B. CAPACITY LOSS

611 Causes

C. INTERNAL SHORTING

621 Detection
622 Verification
623 Action
624 Trouble shooting

8. SAFETY PRECAUTIONS

A. SAFETY EQUIPMENT

801 General
802 Fire extinguisher
803 Face shield or goggles

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A. SAFETY EQUIPMENT (continued)

805 Insulated tools
806 Vaseline or petrolatum
807 First aid materials
809 Rags or toweling

B. HAZARDS

811 Electrical
812 Hydrogen
813 Electrolyte
815 Mercury
816 Hot short

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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